
TrajWiki: Source-Grounded Memory Trajectories for Long-Horizon Dialogue Agents

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Abstract

1 Large language model agents have shown strong capabilities in generating co-
2 herent and contextually appropriate responses, yet robust long-horizon dialogue
3 remains limited by the lack of external memory that is traceable, updatable, and
4 diagnostically transparent. Existing memory-augmented agents often store mem-
5 ories as isolated records or overwriteable states, making it difficult to preserve
6 how information originates, evolves, conflicts, or becomes obsolete over time.
7 We propose **TrajWiki**, a trajectory-based memory framework for long-horizon
8 conversational agents. Instead of treating memory as static entries, TrajWiki rep-
9 resents each memory as a source-grounded evolution trajectory, maintained through
10 immutable episodic snapshots and claim-level operations such as ADD, REVISE,
11 and DEPRECATE. To reduce fragmentation and retrieval cost, TrajWiki further
12 introduces **Memory Wiki**, a persistent intermediate layer that incrementally com-
13 piles dialogue history into structured and interlinked wiki pages capturing salient
14 entities, events, quantities, topics, and conflicts. At inference time, queries are
15 routed hierarchically from relevant wiki pages to linked memory trajectories, then
16 to corresponding snapshots and source messages for evidence-grounded answer
17 synthesis. Experiments on LoCoMo and MedMT show that TrajWiki improves
18 long-horizon dialogue performance across both open-source and closed-source
19 LLM backbones, while providing greater interpretability and diagnostic visibility
20 into memory evolution, retrieval failures, and answer generation. Code is available
21 at <https://anonymous.4open.science/r/TrajWiki-F0EC/>.

22 1 Introduction

23 Large language model (LLM) agents are systems built around large language models that au-
24 tonomously plan, reason, and act by iteratively interacting with users, external tools, and environments
25 to solve complex, multi-step tasks, and they have shown early success in domains such as coding,
26 research assistance, and broader task automation [1–8]. These systems rely on maintaining and
27 updating context across successive interaction steps to support coherent decision-making; however,
28 sustaining coherent long-horizon interaction remains difficult because the underlying models are
29 constrained by finite context windows [9–11]. This limitation becomes particularly acute in long-term
30 dialogue, where useful responses often depend on user-specific facts, evolving preferences, prior
31 commitments, temporally distant events, and evidence scattered across multiple sessions [12–17].
32 Simply extending the prompt with more history is costly and brittle, and does not provide reliable
33 mechanisms for updating, organizing, or diagnosing long-term memory [18–22].

34 Recent work addresses this limitation by equipping LLM agents with explicit external memory.
35 Most memory-augmented systems follow a write/read paradigm: dialogue history is compressed
36 into memory units such as notes, facts, user profiles, or event records, and relevant units are later
37 retrieved to support generation [23]. Other approaches improve memory organization through

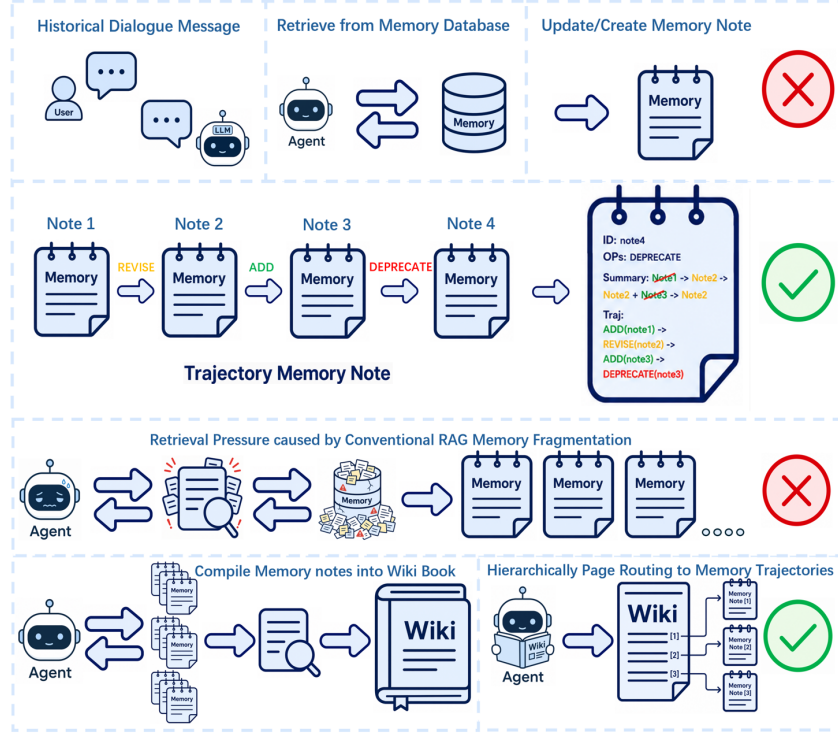


Figure 1: Comparison of conventional agent memory and TrajWiki’s source-grounded trajectory memory and Memory Wiki. Conventional memory retains only the current belief and retrieves flat, isolated memories. TrajWiki preserves source-grounded update trajectories and structures them into a Memory Wiki, enabling reasoning over how facts evolve, conflict, and relate across long-horizon dialogue.

adaptive structures, dynamic linking, or graph-based retrieval [22]. While these methods demonstrate the value of persistent memory, we argue that two challenges remain underexplored, as shown in Figure 1. First, existing systems often retain only the current memory state without explicitly preserving its source-grounded update history. A retrieved memory may state what the agent currently believes, but not how that belief was formed, which claims were revised or deprecated, or which source utterances justify each update. Second, as fine-grained memories accumulate, flat retrieval over isolated memory items becomes increasingly fragmented, especially for temporal, multi-hop, or update-sensitive questions. Conventional retrieval-augmented generation treats knowledge mainly as searchable passages [24], whereas long-horizon dialogue memory also requires modeling how facts evolve, conflict, and relate over time.

In this paper, we propose **TrajWiki**, an interlinked, wiki-style, trajectory-based memory framework for long-horizon conversational agents. TrajWiki addresses the two challenges above through two key mechanisms. First, rather than representing memory as isolated and static records, it models each memory as a source-grounded evolutionary trajectory in which newly generated dialogue segments are stored as immutable episodic snapshots and linked to prior memory states through claim-level operations. Second, instead of repeatedly retrieving relevant information from fragmented raw chunks at query time, it reduces memory fragmentation by organizing trajectories into structured and interlinked wiki pages centered on entities, topics, events, and discrete factual units. At inference time, TrajWiki performs hierarchical retrieval: a query is first routed to relevant wiki pages, then to associated memory trajectories, and finally to episodic snapshots and original source messages, enabling evidence-grounded response generation. In summary, our contributions are fourfold:

- We propose a source-grounded memory trajectory model that records long-term memory evolution through immutable episodic snapshots and claim-level update operations, preserving both the current effective memory and the historical path through which it was produced and making memory evolution inspectable at the level of claims and supporting evidence.

- We introduce an intermediate knowledge organization layer, **Memory Wiki**, that compiles trajectories into structured and interlinked wiki pages for more diagnosable retrieval, while preserving links to trajectories and source messages.
- We further introduce a hierarchical retrieval mechanism over interlinked memory trajectories that separates semantic organization, temporal memory evolution, and fine-grained evidence verification, reducing repeated reasoning over raw dialogue history while retaining provenance for diagnosis.
- We evaluate TrajWiki on LoCoMo and MedMT-Bench [16, 25], covering both open-domain very long-term conversation and domain-specific long multi-turn medical dialogue. Experiments with open-source and closed-source LLM backbones show that TrajWiki improves long-horizon dialogue performance over strong memory-augmented baselines and provides richer diagnostic visibility into memory evolution, retrieval failures, and answer generation.

2 Related Work

2.1 Agent Memory Systems

External memory has become a central mechanism for LLM-based agents that operate beyond a fixed context window. Existing systems have explored multiple ways to store, update, and retrieve long-term interaction history. MemoryBank maintains user-specific memories with time- and importance-aware reinforcement, while MemGPT and MemoryOS frame memory management as a systems problem with hierarchical storage and context control [19, 20, 26]. More recent work introduces richer memory structures, including hierarchical memory trees, temporal knowledge graphs, dynamically linked memory notes, and scalable conversational memory consolidation [21, 22, 27, 28]. In long-horizon dialogue, timeline-based memory, reflective memory management, and graph-structured associative retrieval further demonstrate the value of structured memory for personalization and cross-session response generation [29–31]. Despite these advances, most prior systems focus primarily on what to store and how to retrieve it, rather than explicitly modeling how memory states evolve. Even when memories are updated or linked, the update process is rarely represented as a provenance-aware sequence of immutable snapshots with claim-level edit semantics. This makes it difficult to inspect which claims were added, revised, or deprecated, and which source interactions justified each change.

2.2 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) grounds language generation in external non-parametric memory by retrieving relevant evidence and conditioning generation on the retrieved content [24]. While standard RAG often retrieves flat top- k passages from relatively static corpora, recent methods improve retrieval with additional structure and multi-stage reasoning. Iter-RetGen interleaves retrieval and generation to refine evidence iteratively; RAPTOR organizes documents into a tree of recursive summaries; and GraphRAG builds graph-based indices to support global reasoning and query-focused summarization over large corpora [32–34]. TrajWiki shares the broader insight that retrieval benefits from structure, but targets a different setting. Existing RAG systems are mainly designed to query external document collections, whereas long-horizon dialogue agents must maintain an internal memory that evolves through ongoing interaction. In this setting, relevant evidence depends not only on semantic similarity, but also on when a memory was created, how it was revised, which earlier claims were superseded, and which source messages support the current state.

3 Methodology

We propose **TrajWiki**, as shown in Figure 2, a long-term memory framework for dialogue agents that represents memory as source-grounded trajectories and organizes them into a persistent Memory Wiki. Unlike standard retrieval-augmented generation, which retrieves external passages at inference time [24], TrajWiki first converts dialogue history into evolving memory trajectories, then compiles them into wiki-style pages for hierarchical retrieval and evidence-grounded answer generation.

Given a dialogue history

$$\mathcal{D} = \{m_1, \dots, m_N\}, \quad m_i = (x_i, u_i, \tau_i, \rho_i), \quad (1)$$

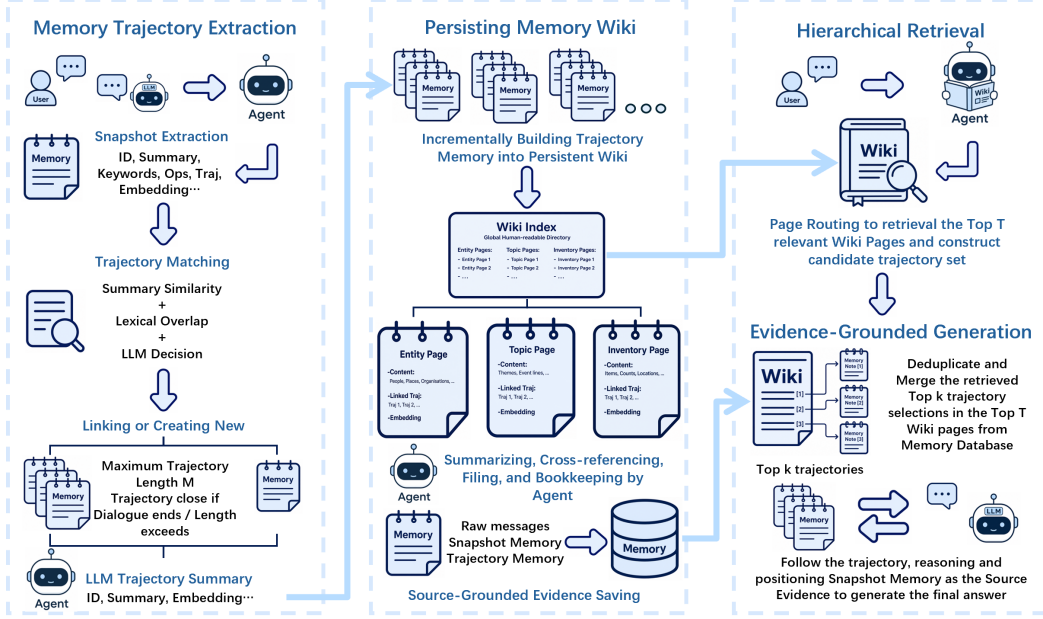


Figure 2: Overview of the TrajWiki architecture. TrajWiki converts dialogue history into source-grounded memory trajectories, compiles them into a persistent and interlinked Memory Wiki, retrieves evidence hierarchically from wiki pages to trajectories and source messages, and generates answers grounded in the retrieved evidence. The Memory Wiki serves as a compounding knowledge artifact, maintaining cross-references, flagged conflicts, and synthesized evidence as memory accumulates over time.

where x_i , u_i , τ_i , and ρ_i denote message content, speaker, timestamp, and source reference, respectively, TrajWiki follows the pipeline

$$\mathcal{D} \rightarrow \Sigma \rightarrow \mathcal{T} \rightarrow \mathcal{W} \rightarrow \mathcal{E}_q \rightarrow a_q. \quad (2)$$

Here, Σ denotes episodic snapshots, $\mathcal{T}$ memory trajectories, $\mathcal{W}$ the Memory Wiki, $\mathcal{E}_q$ the retrieved evidence for query q , and a_q the final answer. This design separates provenance tracking, persistent knowledge organization, and query-time grounding.

3.1 Provenance-Aware Memory Trajectories

TrajWiki stores memory as source-linked trajectories rather than isolated records. Each dialogue exchange is converted into an immutable episodic snapshot containing extracted claims, speaker and temporal context, and links to raw source messages. A memory trajectory is an ordered append-only sequence:

$$T_j = (\sigma_{j,1}, \sigma_{j,2}, \dots, \sigma_{j,n_j}), \quad (3)$$

where each snapshot $\sigma_{j,t}$ remains traceable to the messages that produced it.

For a new snapshot σ , TrajWiki either appends it to the most compatible trajectory or initializes a new one:

$$T^* = \arg \max_{T_j \in \mathcal{T}} g(\sigma, T_j), \quad \sigma \mapsto \begin{cases} T^*, & g(\sigma, T^*) \geq \delta, \\ T_{\text{new}}, & \text{otherwise.} \end{cases} \quad (4)$$

The compatibility function g considers entity continuity, facet overlap, lexical evidence, temporal compatibility, and semantic drift, allowing related updates to be connected without collapsing unrelated facts into overly broad trajectories.

Within each trajectory, memory evolution is represented through claim-level edit operations:

$$C_{j,t} = \text{Apply}(C_{j,t-1}, \mathcal{O}_{j,t}), \quad \mathcal{O}_{j,t} \subseteq \{\text{ADD, REVISE, DEPRECATE}\}. \quad (5)$$

Thus, revised or deprecated claims are not deleted; they remain linked to their original snapshots and source references, making memory updates inspectable over time.

131 3.2 Persistent Memory Wiki

132 Although trajectories preserve provenance, directly retrieving over all snapshots and claims can
 133 become fragmented as memory grows. TrajWiki therefore compiles trajectories into a persistent
 134 **Memory Wiki**, which serves as a readable and routable organization layer between raw memory and
 135 query-time retrieval.

136 Each wiki page is represented as

$$p = (\eta, y, h, L_p), \quad L_p \subseteq \mathcal{T}, \quad (6)$$

137 where η is the page type, y is the title, h is the page content, and L_p is the set of linked trajectories.
 138 We use four page types:

$$\eta \in \{\text{index, entity, topic, inventory}\}.$$

139 The **index** page provides a global directory, **entity** pages organize recurring entities, **topic** pages
 140 capture events or thematic clusters, and **inventory** pages collect list-like or countable facts.

141 Wiki construction is summarized as

$$\mathcal{W} = \text{Compile}(\mathcal{T}; C^{\text{active}}, C^{\text{hist}}, A), \quad (7)$$

142 where C^{active} and C^{hist} denote active and historical claims, and A denotes source anchors and
 143 metadata. The wiki does not replace source evidence with summaries; instead, it organizes accumu-
 144 lated memory while preserving links to trajectories and raw messages. To avoid making trajectories
 145 reachable only through the global directory, TrajWiki encourages non-index coverage:

$$\bigcup_{p \in \mathcal{W}_{\text{non-index}}} L_p \approx \mathcal{T}. \quad (8)$$

146 3.3 Hierarchical Retrieval and Evidence-Grounded Generation

147 Given a query q , TrajWiki retrieves evidence through a hierarchy of wiki pages, trajectories, and
 148 source-linked snapshots. It first routes the query to relevant wiki pages and uses their links to construct
 149 a candidate trajectory set:

$$\mathcal{P}_q = \text{Top}_t\{S_p(q, p) : p \in \mathcal{W}\}, \quad \mathcal{C}_q = \bigcup_{p \in \mathcal{P}_q} L_p. \quad (9)$$

150 Here, S_p combines semantic, lexical, entity, facet, and temporal signals. This stage narrows retrieval
 151 from the entire memory store to organized memory regions.

152 From $\mathcal{C}_q$, TrajWiki selects trajectories using a relevance-and-coverage objective:

$$\mathcal{R}_q = \text{Top}_k\{b(q, T) + \alpha \Delta_{\text{cov}}(T) - \lambda \text{red}(T) : T \in \mathcal{C}_q\}. \quad (10)$$

153 The base score $b(q, T)$ measures relevance, Δ_{cov} rewards complementary entities, events, facets, or
 154 countable items, and red penalizes redundancy. This encourages retrieval of multiple complementary
 155 trajectories for list, count, temporal, and multi-hop questions.

156 Selected trajectories are expanded into provenance-linked evidence:

$$\mathcal{E}_q = \{(T, \sigma, c, \rho, \tau) : T \in \mathcal{R}_q, \sigma \in T, c \in C_T\}. \quad (11)$$

157 Thus, retrieved evidence contains not only text fragments, but also their trajectory, claim, source
 158 reference, and timestamp.

159 The final answer is generated under a source-support constraint:

$$z_q = \text{Synth}_\theta(q, \mathcal{E}_q), \quad a_q = \text{Answer}(z_q), \quad \text{refs}(a_q) \subseteq \text{refs}(\mathcal{E}_q). \quad (12)$$

160 The synthesis z_q records the answerability decision, supporting facts, source references, temporal
 161 anchors, and uncertainties. If the retrieved evidence is insufficient, TrajWiki abstains or performs a
 162 controlled retry by generating retrieval hints and rerouting through the wiki and trajectory layers. In
 163 all cases, the final answer must remain grounded in retrieved source evidence.

164 Overall, TrajWiki decomposes long-term dialogue memory into three layers: source-grounded
 165 trajectories for memory evolution, Memory Wiki pages for persistent organization, and hierarchical
 166 retrieval for query-time grounding. This enables agents to accumulate, revise, and query long-term
 167 memory while preserving the provenance needed to audit both memory updates and generated
 168 answers. We provide the reproducible implementation details, including deterministic scoring
 169 features, structured-output schemas, hyperparameters, validation rules, and retry/repair procedures,
 170 in Appendix A.

Table 1: Overall performance comparison on LoCoMo and MedMT-Bench. For LoCoMo, we report F1, BLEU, and Acc separately for each question type. For MedMT-Bench, we report Acc for each subset. The best performance is marked in bold.

Model	Method	LoCoMo												MedMT-Bench		
		Multi Hop			Temporal			Open Domain			Single Hop			LCMU	RCI	IC
		F1	BLEU	Acc	F1	BLEU	Acc	F1	BLEU	Acc	F1	BLEU	Acc	Acc	Acc	Acc
GPT-4o-mini	Full Context	28.11	21.71	32.98	23.11	16.12	30.22	16.76	13.57	35.42	43.77	36.43	71.72	27.12	11.16	7.47
	Naive RAG	21.78	17.08	20.21	20.35	15.37	24.92	14.89	11.67	26.04	40.49	34.13	60.52	16.43	18.86	7.82
	LangMem	22.01	16.92	28.01	20.59	14.43	31.78	10.31	8.09	23.96	34.4	26.62	55.28	17.78	16.34	8.26
	A-MEM	27.02	20.09	37.58	41.52	35.6	26.17	12.44	9.97	23.96	38.74	34.03	50.54	12.51	17.12	9.33
	Mem0	34.52	25.13	40.07	44.73	37.86	30.56	13.12	9.5	25.02	37.19	33.64	44.48	17.11	14.52	12.17
	TrajWiki	35.23	36.47	59.44	42.42	41.14	35.39	20.56	18.66	42.12	48.72	43.14	78.56	29.33	21.42	17.68
Qwen3-32B	Full Context	27.79	20.91	49.65	32.47	26.56	32.27	15.52	11.44	39.67	29.53	32.71	79.68	29.63	13.16	9.42
	Naive RAG	24.22	19.01	35.82	22.61	18.21	20.25	12.88	8.7	36.46	35.19	27.27	69.08	19.63	21.05	7.62
	LangMem	26.01	24.37	38.16	24.9	23.81	20.28	9.15	6.87	22.92	36.61	34.88	44.15	18.18	16.67	7.74
	A-MEM	25.17	19.12	41.48	29.79	24.63	28.78	11.29	7.68	41.67	33.76	26.21	63.26	13.89	17.07	10.36
	Mem0	26.32	20.17	45.74	32.41	29.68	31.49	15.03	11.28	42.7	42.58	35.15	73.72	17.89	23.68	11.21
	TrajWiki	31.42	30.56	50.33	33.48	31.26	37.33	17.86	16.44	39.68	41.88	40.42	78.34	33.42	27.33	15.34
Qwen3-8B	Full Context	39.05	31.16	39.36	33.64	27.19	26.17	20.67	16.52	31.54	43.27	43.45	78.21	26.69	10.22	6.76
	Naive RAG	27.25	21.05	29.79	27.86	22.96	18.69	12.18	8.98	38.54	46.32	41.44	64.8	16.32	18.32	6.42
	LangMem	16.48	15.08	27.09	17.85	15.83	15.61	9.47	6.96	24.58	36.24	35.16	51.77	17.81	15.47	7.74
	A-MEM	27.54	22.08	35.46	34.30	28.38	21.18	11.39	8.62	22.29	42.3	37.66	59.45	12.04	16.32	8.76
	Mem0	31.73	24.82	38.27	28.96	26.24	20.16	15.03	11.28	21.18	42.58	35.15	57.64	15.42	13.62	10.52
	TrajWiki	30.25	32.14	42.14	31.48	30.48	34.62	16.89	13.52	39.78	43.46	37.24	63.48	27.56	20.11	14.28

4 Experiment

4.1 Datasets and Evaluation

We evaluate **TrajWiki** on two long-horizon dialogue benchmarks. Our primary benchmark is LoCoMo, a long-term conversational memory dataset with gold question-answer pairs [16]. We use its QA task, which covers *single-hop*, *multi-hop* requiring cross-session evidence aggregation, *temporal reasoning*, and *open-domain / world-knowledge* questions. Following Mem0 [21], we exclude the adversarial category to ensure fair comparison with prior memory-augmented baselines. To assess robustness beyond open-domain conversations, we further use MedMT-Bench as a domain-specific stress test [25]. MedMT-Bench evaluates long multi-turn medical conversations using instance-level rubrics rather than a single reference answer. Since our goal is to evaluate memory mechanisms rather than medical knowledge or multimodal reasoning, we select memory-relevant cases from three categories: *Long-Context Memory and Understanding (LCMU)*, *Resistance to Contextual Interference (RCI)*, and the *Information Contradiction (IC)* subset of *Instruction Clarification*. These categories test long-history alignment, robustness to contextual noise, and conflict detection between historical facts and current user statements.

We compare TrajWiki with five baselines: **Full Context**, which directly conditions on the complete dialogue history; **Naive RAG**, evaluated with both session-level and turn-level retrieval granularity [24]; **LangMem**, which extracts salient conversational information for persistent memory retrieval [35]; **A-MEM**, which dynamically links structured memory notes [22]; and **Mem0**, which performs incremental memory extraction, consolidation, and retrieval [21]. For LoCoMo, we report F1 and BLEU-1 as lexical overlap metrics [36]. Since long-horizon dialogue answers may be semantically correct despite different surface forms, we additionally use a strict LLM-as-a-Judge protocol to evaluate answer correctness [37]. For MedMT-Bench, we follow its rubric-based setting and report test-point pass accuracy. Details of metric computation are provided in Appendix B.1.

4.2 Implementation Details

For all baselines, we follow their original configurations, hyperparameters, and system prompts. When details are underspecified, we use the closest publicly described setting while keeping the backbone model, answering format, and evaluation protocol consistent across methods. For closed-source experiments, we use GPT-4o-mini as the backbone model and LLM judge. For open-source experiments, we use Qwen3-8B and Qwen3-32B in non-thinking mode as backbone models, and Qwen3-8B as the judge. TrajWiki uses Qwen3-Embedding-8B for all embedding-based retrieval

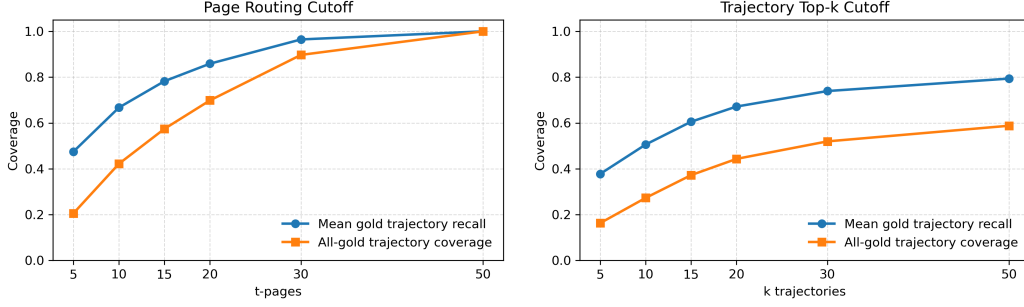


Figure 3: Hyperparameter Retrieval Cutoff Analysis on the LoCoMo. The left plot varies the wiki page routing cutoff t , showing how increasing routed pages improves gold trajectory coverage in the page-induced candidate universe. The right plot varies trajectory top- k , showing that selecting more trajectories improves recall but increases the evidence passed to answer generation.

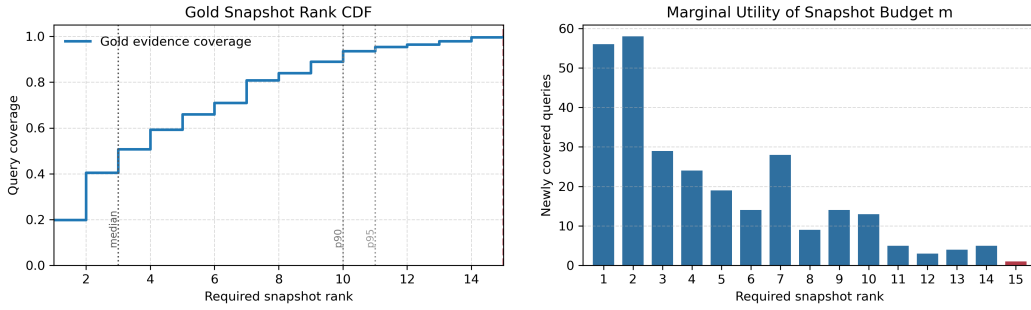


Figure 4: Snapshot Budget Analysis for trajectory length m on the LoCoMo. The left plot shows the cumulative fraction of queries whose gold evidence is covered within a given required snapshot rank. The right plot shows the marginal number of newly covered queries at each rank, indicating that most gains occur at shallow depths while $m = 15$ covers the remaining long-tail evidence.

components. We report all major prompts in Appendix C. For TrajWiki, the default setting uses maximum trajectory length $m = 15$, routes each query through the top $t = 15$ wiki pages, and selects the top $k = 15$ trajectories. These settings balance retrieval coverage and computational cost.

4.3 Empirical Results

Table 1 reports the main results on LoCoMo and MedMT-Bench [16, 25]. Across both benchmarks and both GPT- and Qwen-based backbones, **TrajWiki** consistently outperforms most of the baselines, including Full Context, Naive RAG, LangMem, A-MEM, and Mem0 [21, 22, 35]. On LoCoMo, TrajWiki achieves better performance and improves over the strongest memory baseline, Mem0. The gains are most pronounced on multi-hop questions, where answering requires aggregating evidence distributed across sessions. This suggests that source-grounded trajectories and Memory Wiki routing help recover complementary memory traces beyond local semantic similarity. TrajWiki also shows a consistent advantage on temporal reasoning questions, although the margin is smaller, reflecting the difficulty of resolving temporally distant and evolving dialogue evidence. On MedMT-Bench, the performance gap becomes more substantial. Most baselines degrade in this domain-specific setting, especially on the Information Contradiction subset, where models must detect conflicts between historical facts and the current user statement. In contrast, TrajWiki remains robust across the three medical scenarios, demonstrating that its memory mechanism transfers beyond open-domain conversations to high-constraint professional dialogues. We attribute this improvement to two aspects of the design: trajectory-level memory preserves how claims evolve over time, while wiki-based routing reduces retrieval fragmentation before evidence-level expansion.

4.4 Hyperparameter Analysis

We analyze three key hyperparameters of TrajWiki on LoCoMo: the maximum trajectory length m , the number of routed wiki pages t , and the number of selected trajectories k . Rather than

Table 2: Counterfactual ablation on LoCoMo with GPT-4o-mini. All counterfactual variants use $k = 15$ and a 32K offline context budget without rerunning answer generation. “Unsup. Risk” is an unsupported-evidence risk proxy.

Setting	Removed / Restricted Component	Cand. Univ.	Gold Ref Cov.	Gold Traj. R@15	Ctx. Tokens	Unsup. Risk
Full TrajWiki	None	55.6	0.610	0.610	2.7K	0.645
Direct Trajectory Retrieval	Memory Wiki routing	130.4	0.356	0.346	25.6K	0.872
Wiki Summaries Only	Source-grounded trajectory expansion	41.2	0.000	0.807	25.3K	1.000
Flat Raw-Memory Retrieval	Trajectory and wiki organization	594.7	0.237	0.457	0.7K	0.918
Latest Snapshot Only	Historical snapshot depth	15.0	0.613	0.610	6.4K	0.649

rerunning answer generation for every setting, we use the compact retrieval diagnostics saved during benchmarking to perform an offline coverage analysis. Figure 3 compares the effects of page routing cutoff and trajectory top- k . Increasing t substantially improves the probability that gold trajectories enter the page-induced candidate universe: small values such as $t = 5$ or $t = 10$ miss many supporting trajectories, while $t = 15$ provides a practical trade-off and $t = 20$ further improves coverage at the cost of a larger candidate set. Increasing k similarly improves gold trajectory coverage, but also increases the amount of evidence passed to answer generation, which may introduce additional noise. Figure 4 analyzes the required snapshot depth inside gold trajectories. Most queries require only shallow trajectory history, while a small long tail requires deeper snapshots. The marginal utility plot shows that most additional coverage is gained at small ranks, with diminishing returns after $m = 10$, while $m = 15$ covers the remaining long-tail evidence. These results support the design of TrajWiki as a hierarchical memory system: wiki routing controls retrieval breadth, trajectory selection controls evidence depth, and bounded trajectories preserve sufficient historical context without requiring unbounded memory expansion.

4.5 Counterfactual Ablation

We conduct a counterfactual ablation on the LoCoMo multi-hop split using GPT-4o-mini. The goal is to isolate how TrajWiki’s memory organization affects evidence retrieval, grounding, and context efficiency. Since the counterfactual variants are derived from a single full TrajWiki run and do not rerun answer generation, we report retrieval and context-grounding metrics for the ablated variants; observed answer-level metrics are available only for the full TrajWiki run. A more detailed budget-controlled analysis is provided in Appendix B.2. Table 2 shows the main results at $k = 15$ under a 32K offline context budget. Removing the Memory Wiki and directly ranking trajectories increases the candidate universe from 55.6 to 130.4 trajectories, while reducing gold source-reference coverage from 0.610 to 0.356. This suggests that wiki routing provides useful semantic organization rather than acting as a simple lossy filter. The wiki-only setting further shows that summaries alone are insufficient for source-grounded QA: although they can point to relevant trajectories, they do not recover gold source references. Flat raw-memory retrieval also has a much larger candidate universe and higher unsupported-evidence risk. Overall, the results support TrajWiki’s combination of persistent wiki routing, source-grounded trajectories, and snapshot-level evidence expansion.

4.6 Trajectory Semantic Drift Analysis

We further analyze whether memory trajectories remain semantically coherent as they accumulate updates over time. For each trajectory, we compute the cosine similarity between the embeddings of its first and latest snapshots, which measures the semantic span of the trajectory, and the average cosine similarity between adjacent snapshots, which measures local continuity during memory updates. This analysis is performed offline using the saved snapshot embeddings from the LoCoMo run and does not involve additional model calls. As shown in Figure 5, the left panel indicates that most non-singleton trajectories maintain moderate to high head-tail similarity, while only a small fraction falls into the possible-drift region. The right panel further shows that longer trajectories naturally cover a broader semantic range, leading to lower head-tail similarity, but their adjacent-update similarity remains relatively stable. This suggests that TrajWiki trajectories support gradual memory evolution rather than arbitrary aggregation: even when a trajectory spans multiple related events, consecutive updates remain locally coherent. These results support the design of provenance-aware trajectories as a bounded yet extensible unit for long-term memory, where semantic drift can be measured and audited rather than hidden inside overwritten memory states.

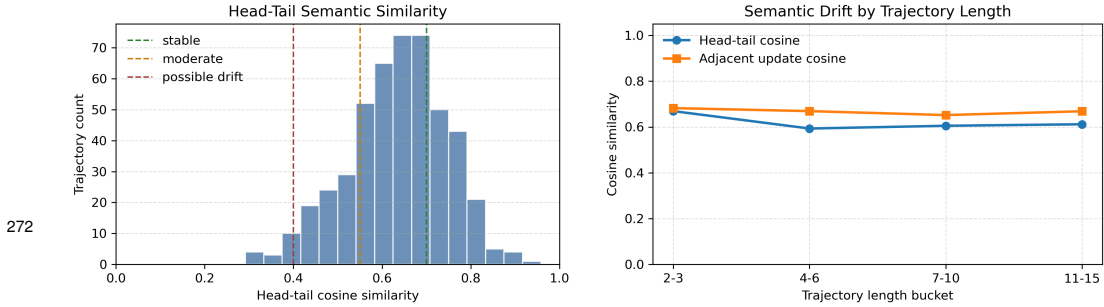


Figure 5: Trajectory Analysis. The left panel shows head-tail semantic similarity for non-singleton trajectories, while the right panel breaks semantic drift down by trajectory length and contrasts global head-tail span with local adjacent-snapshot similarity. Most trajectories remain semantically coherent, and longer trajectories mainly increase global span rather than causing unstable local updates.

4.7 Computational Cost and Scalability Analysis

We analyze the observed cost of TrajWiki on the LoCoMo using GPT-4o-mini. The cost is concentrated in memory construction and evidence organization rather than final answer generation: excluding benchmark-only evaluation, the run uses 43.28M deployment tokens, including 23.71M for reusable memory construction, 10.84M for query-time retrieval, and 6.92M for answer generation. This reflects TrajWiki’s main trade-off: additional computation is used to build source-grounded trajectories and a persistent Memory Wiki. The scalability analysis further shows that the Memory Wiki reduces the query-time trajectory search space. Across dialogue samples, direct retrieval would consider 130.4 trajectories per query on average, while wiki routing reduces this to 55.6 candidate trajectories, a $2.35\times$ reduction. Detailed phase-level costs, latency breakdowns, and memory-scaling results are provided in Appendix B.3.

4.8 Diagnostic Visibility and Limitations

TrajWiki improves auditability by preserving source links across snapshots, trajectories, wiki pages, retrieval evidence, and final answers, but provenance-aware does not mean provenance-perfect. The system still depends on structured LLM outputs for extraction, memory updating, wiki construction, answer synthesis, and validation, so malformed outputs can trigger costly fallback or repair paths, especially for weaker non-instruction-tuned models. It also relies on LLM-generated summaries to organize raw dialogue history; although source links are retained, repeated abstraction can shift the salience of specific facts and cause answer generation to rely on compressed memory rather than the most relevant raw evidence. Finally, our diagnostics expose unsupported-answer risk, invalid supporting references, and likely failure stages, but these are proxy diagnostics rather than human-verified audit accuracy. Thus, TrajWiki should be viewed as an auditable long-term memory framework rather than a fully solved grounding system. Appendix B.4 reports quantitative diagnostic statistics and a case study of summary-mediated information loss.

5 Conclusion

We introduced **TrajWiki**, a long-term memory framework that represents dialogue history as provenance-aware memory trajectories and organizes them into a persistent Wiki-style semantic layer. By separating memory evolution and persistent organization, TrajWiki enables agents to accumulate, revise, and retrieve long-term conversational knowledge while preserving source-level evidence. Our experiments show that hierarchical retrieval over Wiki pages and trajectories improves the ability to locate multi-hop evidence, while evidence-grounded answer generation and post-hoc diagnostics make failures more interpretable. The results also highlight an important trade-off. TrajWiki incurs additional computational cost, but this cost supports reusable semantic organization, provenance tracking, and auditable retrieval, and can be amortized across future queries in long-term agent settings. Overall, TrajWiki demonstrates that treating memory as an evolving, source-grounded structure rather than a flat retrieval pool is a promising direction for building more reliable long-term dialogue agents.

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APPENDIX

A Reproducible Implementation Details

This appendix provides the operational details needed to reproduce the main TrajWiki pipeline. TrajWiki combines deterministic scoring and validation with structured LLM calls. We therefore distinguish between deterministic candidate construction, structured model decisions, and fallback rules.

A.1 Memory construction.

Dialogue messages are first grouped into exchange-level inputs. For each exchange, TrajWiki asks the backbone model to produce an episodic memory seed with three fields: `summary_content`, `context`, and `keywords`. A separate claim extraction stage then emits source-grounded atomic claims:

$$c = (\text{text}, \text{status}, \text{source_message_ids}, \text{supporting_quote}),$$

where $\text{status} \in \{\text{active}, \text{deprecated}, \text{contradictory}, \text{needs-confirmation}\}$. All claims are validated against the source messages. Unsupported source ids, empty claims, and ungrounded exact terms are discarded. If the claim extractor fails or omits high-confidence source details, a deterministic preservation fallback extracts must-preserve surfaces such as names, places, books, instruments, activities, counts, temporal expressions, and quoted titles from the raw exchange.

A.2 Trajectory matching.

For each new episodic snapshot σ , TrajWiki first builds a deterministic feature vector from the snapshot text, extracted claims, and linked source messages. The feature set includes:

$$K_\sigma, E_\sigma, X_\sigma, F_\sigma,$$

where K_σ are lexical keywords, E_σ are entity mentions, X_σ are exact source-backed terms, and F_σ are relation/value facets such as `home_country`, `activity_location`, `research_topic`, and `event_type`. For each open trajectory T , TrajWiki stores analogous trajectory-level signals from active claims, historical item terms, retrieval summaries, and the latest snapshot.

Candidate trajectories are prefiltered by keyword/entity/facet overlap. If any overlap exists, only overlapping trajectories are kept, capped at 32 candidates by overlap count; otherwise all open trajectories are allowed as a fallback. Each candidate is scored by

$$g_{\text{det}}(\sigma, T) = 0.60 \cos(e_\sigma, e_T^{\text{sum}}) + 0.20 \cos(e_\sigma, e_T^{\text{latest}}) + 0.20 J(K_\sigma, K_T) \\ + b_{\text{entity}} + b_{\text{facet}} + b_{\text{cont}} - p_{\text{mismatch}}.$$

Here J is keyword Jaccard overlap. The entity bonus is 0.08 when normalized entity keys overlap. The facet bonuses are 0.06 for exact facet-value overlap and 0.03 for facet-tag overlap. The continuity bonus is

$$b_{\text{cont}} = \min(0.14, 0.04 |\text{specific}(\sigma) \cap \text{specific}(T)|).$$

A mismatch penalty of 0.10 is applied when a candidate shares only a broad entity but no specific continuity terms; a smaller penalty of 0.04 is applied when lexical overlap exists without specific continuity.

The top three candidates under g_{det} are then passed to a structured `trajectory_match` decision:

$$\text{decision} \in \{\text{CONTINUE}, \text{NEW}\}, \quad \text{selected_candidate} \in \{T1, T2, T3, \text{null}\}.$$

If structured output is unavailable, TrajWiki falls back to a text parser. In mock deterministic mode only, the best candidate is continued when $g_{\text{det}} \geq \delta$, with $\delta = 0.72$. Thus, δ is a deterministic fallback threshold rather than the sole decision rule in remote-model experiments.

A.3 Claim evolution.

Within a trajectory, TrajWiki maintains an append-only claim lifecycle. Exact duplicate claims are first matched by normalized text. If the same claim text changes status, TrajWiki emits a status

operation: DEPRECATE when the new status is deprecated, otherwise REVISE. For unmatched new claims, previous claims are shortlisted by keyword overlap, keeping at most three candidate previous claims. A structured `claim_transition_judge` call then decides

$$\text{decision} \in \{\text{REVISE}, \text{ADD}\}.$$

A REVISE operation creates a new claim and marks the selected previous claim as deprecated; an ADD operation appends the claim without replacing prior evidence. If the judge fails or no previous claim candidate exists, TrajWiki falls back to ADD. Model-suggested operations are treated as hints only; the persisted state is derived by the validated deterministic procedure above.

A.4 Memory Wiki compilation.

After replaying a sample’s dialogue into trajectories, TrajWiki compiles a sample-level Memory Wiki from the current trajectory store. In the implementation used for our experiments, wiki pages are refreshed after memory replay rather than edited one page at a time online. This preserves stable trajectory, snapshot, claim, and source ids while allowing page organization to be recomputed from the latest memory state.

The compiler first creates candidate page seeds:

$$\eta \in \{\text{index}, \text{entity}, \text{topic}, \text{inventory}\}.$$

The `index` page links all trajectories. `entity` pages group trajectories sharing an entity. `inventory` pages are created for list-like trajectories with multiple exact terms, display items, counts, or inventory-like facets. `topic` pages group repeated retrieval keywords. Broad entity pages are split into entity-facet pages when they exceed a trajectory threshold. Seeds with more than six trajectories are split into medium-granularity shards using trajectory-summary embeddings; the target page size is four trajectories and the maximum non-index page size is six. Redundant topic seeds are suppressed when at least 80% of their trajectories are already covered by entity or inventory seeds. A coverage audit then adds rescue pages for any trajectory reachable only through the index page.

Each page is compiled into markdown with required sections: overview, key facts, items/counts, linked trajectories, and conflicts/uncertainty. If the LLM page compiler returns empty text, missing sections, or placeholder-only content, TrajWiki falls back to deterministic markdown generated from linked trajectory summaries and evidence cards. Each page also stores a routing text, keywords, linked trajectory ids, page type, source-backed terms, display items, counts, and a page embedding.

A.5 Hierarchical retrieval.

Given a query q , TrajWiki first classifies the query shape using deterministic lexical rules: list-like, count-like, multi-entity, comparison-like, duration-count, and item family. It then embeds the query and routes it to wiki pages. Index pages are suppressed when non-index pages exist, so the wiki directory does not dominate retrieval.

Each page receives dense and sparse scores. The dense score combines page embedding similarity with entity, reflection, granularity, and answer-family adjustments:

$$s_p^{\text{dense}} = \cos(e_q, e_p) + b_{\text{entity}} + b_{\text{reflect}} + a_{\text{gran}} + 0.10s_{\text{family}} - 0.12p_{\text{family}}.$$

The sparse score is lexical overlap over page keywords, title, entity keys, exact terms, source-event terms, and family terms:

$$s_p^{\text{sparse}} = J(K_q, K_p) + 0.35s_{\text{family}} - 0.25p_{\text{family}}.$$

Dense and sparse rankings are fused by reciprocal rank fusion:

$$\text{RRF}(x) = \frac{1}{60 + r_{\text{dense}}(x)} + \frac{1}{60 + r_{\text{sparse}}(x)}.$$

The top page candidate pool is then optionally reranked by the backbone model with a structured textual output. If reranking fails, the RRF order is used.

Selected wiki pages induce a candidate trajectory universe:

$$\mathcal{C}_q = \bigcup_{p \in \mathcal{P}_q} L_p.$$

494 Candidate trajectories are scored similarly:

$$\begin{aligned}
s_T^{dense} &= 0.75 \cos(e_q, e_T^{sum}) + 0.15 \cos(e_q, e_T^{latest}) + 0.10 b_{\text{entity}/\text{facet}} \\
&\quad + 0.08 s_{\text{family}} + 0.10 s_{\text{event}} - 0.04 p_{\text{family}}, \\
s_T^{sparse} &= J(K_q, K_T) + 0.60 s_{\text{family}} \\
&\quad + 0.75 s_{\text{event}} - 0.25 p_{\text{family}}.
\end{aligned}$$

495 Dense and sparse trajectory ranks are again fused with RRF. The top trajectory pool is reranked by
496 the backbone model when available.

497 For list, count, comparison, and multi-entity questions, TrajWiki performs coverage-aware greedy
498 selection after reranking. The first selected trajectory is the strongest relevance anchor. Subsequent
499 selections maximize

$$\begin{aligned}
S(T) &= s_{\text{rank}}(T) + 0.90 \Delta_{\text{entity}} + 0.65 \Delta_{\text{facet}} + 0.30 \Delta_{\text{item}} + 0.12 \Delta_{\text{term}} \\
&\quad + 0.18 \mathbb{I}_{\text{inventory}} + 0.18 \mathbb{I}_{\text{count}} + 0.75 s_{\text{family}} - p_{\text{red}},
\end{aligned}$$

500 where p_{red} penalizes repeated entities, repeated terms, repeated item terms, repeated clusters, and
501 answer-family mismatch. For non-coverage queries, TrajWiki simply fills from the reranked order.

502 **Evidence expansion and final answer.** Selected trajectories are expanded into snapshots. TrajWiki
503 first keeps the top snapshot per selected trajectory, then fills the remaining snapshot budget by
504 query-snapshot embedding similarity. The snapshot budget is $2k$. Depending on the expansion
505 mode, neighboring snapshots and update-linked snapshots are also added. Source messages linked to
506 selected snapshots are compacted into a final prompt context containing wiki pages, selected trajectory
507 summaries, snapshots, active claims, source messages, temporal anchors, and conflict/deprecated-
508 claim diagnostics.

509 LoCoMo answer generation uses a structured evidence synthesis schema:

```

can_answer,  answer_type,
final_answer, supporting_facts,
supporting_source_refs,
counted_events,  excluded_events,
uncertainties,  abstain_reason.

```

510 All supporting refs must be visible source refs in the retrieved context. Invalid refs are removed. If
511 an answer has no valid source support after validation, TrajWiki converts it to an abstention. Count
512 answers are additionally validated by removing duplicate, future/planned, uncertain, or unsupported
513 events. Type mismatches trigger a typed retry or safe abstention.

514 A.6 Retry and repair.

515 Retrieval reflection is used only for LoCoMo when the initial answer abstains or retrieval is weak.
516 A retrieval bundle is considered weak if it has no source messages, no candidate trajectories, no
517 selected snapshots, or no active claims in the answer context. Reflection produces a structured query
518 rewrite with target entities, event terms, temporal terms, must-find terms, candidate wiki slugs, and
519 raw search terms. TrajWiki reroutes through the wiki using these hints. If the reflected answer still
520 abstains or the synthesis schema returns `can_answer=false`, a raw-message rescue pass retrieves
521 lexical/semantic raw source candidates without changing the memory state.

522 Post-generation repair is triggered by deterministic checks for unsupported extra list items, unsup-
523 ported counts, missing supported list items, over-generic items, scope-mismatched items, unresolved
524 bridge aliases, or answer-type mismatch. Repairs are generated using only retrieved evidence. A re-
525 paired answer is discarded if it is malformed, loses required supported values, introduces unsupported
526 extras, or fails arbitration; in such cases the initial answer is kept or TrajWiki emits a safe abstention.

527 A.7 Hyperparameters.

528 Unless otherwise stated, the LoCoMo experiments use $m = 15$ memory trajectories/pages exposed to
529 downstream stages, $t_{\text{pages}} = 15$ wiki pages, $k = 15$ selected trajectories, neighbor radius 1, retrieval

expansion mode update_linked_plus_neighbors, trajectory candidate prefilter cap 32, rerank candidate-pool size 12, RRF constant 60, maximum non-index wiki page size 6, target wiki page size 4, and mock-mode trajectory continuation threshold $\delta = 0.72$. The implementation supports remote, local, and OpenAI-compatible providers; structured-output schemas are used when supported, with parser and deterministic fallbacks otherwise.

B Evaluation Details

B.1 Evaluation Metrics

We report two automatic text-based metrics, F1 and BLEU-1, following common practice in long-form question answering benchmarks. In our implementation, both metrics are computed over a canonicalized semantic representation rather than raw surface text. This reduces spurious penalties when an answer expresses the correct fact using longer or slightly different natural-language phrasing.

Given a reference answer y and a model answer $\hat{y}$, we first extract canonical answer slots from both answers. Let

$$\mathcal{C}(y) = \{(s_i, v_i)\}_{i=1}^n, \quad \mathcal{C}(\hat{y}) = \{(\hat{s}_j, \hat{v}_j)\}_{j=1}^m,$$

where s denotes a semantic slot identifier and v denotes a normalized canonical value.

F1. We compute F1 using slot-aware semantic matching between reference and candidate canonical values. Unlike exact set overlap, our current implementation uses a conservative soft matching policy within the same slot. A candidate value may match a reference value if they are identical after normalization, differ only by articles, punctuation, or simple inflection, contain benign modifiers such as “LGBTQ support group” versus “support group”, or match through a predefined alias relation such as “NYC” and “New York City”. Contrastive modifiers, numbers, colors, person names, and other distinguishing terms are preserved to avoid over-matching.

Formally, let

$$\mu((\hat{s}, \hat{v}), (s, v)) \in \{0, 1\}$$

denote whether a candidate slot-value pair matches a reference slot-value pair under this policy. Matching is restricted to pairs with $\hat{s} = s$, and each candidate value can match at most one reference value. We therefore compute a maximum one-to-one matching

$$M^* = \arg \max_M \sum_{((\hat{s}, \hat{v}), (s, v)) \in M} \mu((\hat{s}, \hat{v}), (s, v)).$$

Precision, recall, and F1 are then

$$P = \frac{|M^*|}{|\mathcal{C}(\hat{y})|}, \quad R = \frac{|M^*|}{|\mathcal{C}(y)|},$$

$$F1 = \frac{2PR}{P + R}.$$

If both canonical sets are empty, F1 is defined as 1.0; if only one side is empty, F1 is 0.0. This metric rewards complete semantic coverage of the reference facts, penalizes missing reference facts through recall, and penalizes unsupported extra canonical facts through precision.

BLEU-1. We also report BLEU-1 as clipped unigram precision over canonical answer text. We linearize the canonical slot-value representation into a deterministic text sequence, tokenize it after lowercasing and punctuation normalization, and compute

$$\text{BLEU-1} = \frac{\sum_{w \in V} \min(\text{count}_{\hat{y}}(w), \text{count}_y(w))}{\sum_{w \in V} \text{count}_{\hat{y}}(w)}.$$

Unlike standard BLEU, we do not apply a brevity penalty. BLEU-1 therefore measures how much of the candidate’s canonical unigram content is supported by the reference answer. If both token sequences are empty, BLEU-1 is defined as 1.0; if only one is empty, it is 0.0.

Together, F1 evaluates slot-level semantic fact coverage with conservative soft matching, while BLEU-1 measures canonical unigram precision. These deterministic metrics complement the LLM-based judge by providing reproducible lexical-semantic scores over normalized answer content.

Table 3: Full offline ablation at 32K context budget. All settings use $k = 15$. Counterfactual variants report retrieval/context proxies only.

Setting	Cand. Univ.	Gold Ref Cov.	Gold Traj. R@15	All Ref Rate	Ctx. Tokens	Unsup. Risk
Full TrajWiki	55.6	0.610	0.610	0.355	2.7K	0.645
Direct Trajectory Retrieval	130.4	0.356	0.346	0.128	25.6K	0.872
Wiki Summaries Only	41.2	0.000	0.807	0.000	25.3K	1.000
Flat Raw-Memory Retrieval	594.7	0.237	0.457	0.082	0.7K	0.918
Latest Snapshot Only	15.0	0.613	0.610	0.351	6.4K	0.649
Latest Two Snapshots	22.5	0.564	0.558	0.287	6.8K	0.713
Source-Linked Claims Only	347.0	0.088	0.136	0.011	0.2K	0.989

B.2 Offline Counterfactual Retrieval and Context Ablation

We provide a more detailed offline ablation analysis to complement the compact results in Table 2. The analysis is conducted on the LoCoMo multi-hop split using the same full TrajWiki run with GPT-4o-mini. The counterfactual settings reuse saved retrieval diagnostics, gold evidence labels, trajectory/page rankings, source references, and token estimates. They therefore evaluate evidence availability and context cost, but do not rerun answer generation. Consequently, we do not report true judge accuracy for counterfactual variants.

Metrics. **Candidate universe** is the average number of memory items considered before final selection. **Gold Ref Cov.** measures the fraction of gold source references recovered by the selected context. **Gold Traj. R@15** measures recall of gold trajectories under the top-15 trajectory cutoff. **All Ref Rate** is the fraction of queries for which all gold source references are recovered. **Ctx. Tokens** is an offline whitespace-based context token estimate. **Unsup. Risk** is a proxy for unsupported evidence risk, indicating that selected evidence lacks sufficient source support for the gold references.

Ablation settings. **Full TrajWiki** uses the observed wiki-routed retrieval and source-grounded trajectory expansion. **Direct Trajectory Retrieval** removes wiki routing and ranks all trajectories in the same dialogue sample. **Wiki Summaries Only** keeps wiki pages but does not expand them into source-linked trajectory evidence. **Flat Raw-Memory Retrieval** ranks raw dialogue messages directly using lexical signals. **Latest Snapshot Only** and **Latest Two Snapshots** restrict each selected trajectory to its most recent one or two snapshots. **Source-Linked Claims Only** keeps only claims with explicit source links and is included as a diagnostic proxy rather than a full source-support-constraint ablation.

Table 3 reports all variants at the 32K budget. The full TrajWiki setting provides the best balance between gold source-reference coverage, trajectory recall, and candidate-space size. Direct trajectory retrieval considers more than twice as many candidates but recovers less gold evidence under the same offline selection protocol. Wiki-only retrieval has high trajectory recall but zero gold source-reference coverage, showing that wiki summaries alone are not sufficient as answer evidence. Flat raw-memory retrieval is cheap under this lexical selector but has a very large memory universe and high unsupported-evidence risk. The source-linked-claims-only diagnostic has very low context cost but poor coverage, indicating that source filtering alone cannot replace structured retrieval.

Table 4 reports the budget-controlled results for the main settings. Full TrajWiki reaches its evidence coverage with a compact selected context. In contrast, direct trajectory retrieval improves as the budget increases, but even at 32K it remains below full TrajWiki in gold source-reference coverage and trajectory recall. Wiki-only retrieval increases trajectory recall with larger budgets but continues to have zero gold source-reference coverage, because wiki summaries are not source evidence. This supports the need for both hierarchical routing and source-grounded trajectory expansion.

B.3 Computational Cost and Scalability Analysis

We provide a detailed cost and scalability analysis for TrajWiki on the LoCoMo multi-hop split using GPT-4o-mini, with $m = 15$, $t_{\text{pages}} = 15$, and $k = 15$. Since TrajWiki uses multiple asynchronous workers, we report accumulated provider-side latency rather than end-to-end wall-clock time. We also separate deployment costs from benchmark-only evaluation costs. This section reports observed

Table 4: Budget-controlled offline ablation. Values are shown for $k = 15$. “Ctx.” is estimated context tokens.

Setting / Budget	Gold Ref Cov.	Gold Traj. R@15	Ctx.	Unsup. Risk
Full TrajWiki, 4K	0.606	0.605	2.6K	0.649
Full TrajWiki, 8K	0.610	0.610	2.7K	0.645
Full TrajWiki, 16K	0.610	0.610	2.7K	0.645
Full TrajWiki, 32K	0.610	0.610	2.7K	0.645
Direct Trajectory Retrieval, 4K	0.041	0.041	0.6K	0.993
Direct Trajectory Retrieval, 8K	0.110	0.108	3.2K	0.979
Direct Trajectory Retrieval, 16K	0.216	0.210	9.9K	0.936
Direct Trajectory Retrieval, 32K	0.356	0.346	25.6K	0.872
Wiki Summaries Only, 4K	0.000	0.276	3.3K	1.000
Wiki Summaries Only, 8K	0.000	0.442	7.1K	1.000
Wiki Summaries Only, 16K	0.000	0.656	15.1K	1.000
Wiki Summaries Only, 32K	0.000	0.807	25.3K	1.000
Flat Raw-Memory Retrieval, 4K	0.237	0.457	0.7K	0.918
Flat Raw-Memory Retrieval, 8K	0.237	0.457	0.7K	0.918
Flat Raw-Memory Retrieval, 16K	0.237	0.457	0.7K	0.918
Flat Raw-Memory Retrieval, 32K	0.237	0.457	0.7K	0.918

Table 5: Phase-level cost breakdown for TrajWiki on LoCoMo multi-hop. Tokens and latency are aggregated across the full run. Benchmark evaluation is reported separately from deployment cost.

Phase	Deployment Scope	Reuse Scope	Calls	Tokens	Provider Latency
Memory construction	Deployment	Upfront reusable	2256	23.71M	25.66h
Query-time retrieval	Deployment	Per query	846	10.84M	1.58h
Answer generation	Deployment	Per query	564	6.92M	0.44h
Repair / validation	Deployment	Mixed	768	1.81M	2.50h
Benchmark evaluation	Benchmark only	Benchmark only	846	0.42M	0.45h

609 TrajWiki costs only; we do not claim a token break-even against proxy baselines without answer-level
610 reruns.

611 Table 5 gives the phase-level cost breakdown. The largest source of token usage is memory con-
612 struction, which consumes 23.71M tokens, followed by query-time retrieval with 10.84M tokens and
613 answer generation with 6.92M tokens. This confirms that the dominant cost of TrajWiki is not the
614 final answer itself, but the construction and organization of source-grounded long-term memory.

615 Figure 6 visualizes the same token distribution. Memory construction accounts for 54.3% of all
616 measured tokens, query-time retrieval accounts for 24.8%, answer generation accounts for 15.8%,
617 repair/validation accounts for 4.1%, and benchmark-only evaluation accounts for only 1.0%. Thus,
618 evaluation overhead is not the main driver of the reported cost. Instead, the cost reflects TrajWiki’s
619 design choice to transform raw dialogue into trajectories, claims, snapshots, and wiki pages before
620 answer generation.

621 Figure 7 shows an even stronger concentration in provider-side latency. Memory construction
622 accounts for 83.8% of accumulated provider latency, compared with 8.1% for repair/validation, 5.2%
623 for query-time retrieval, 1.5% for benchmark evaluation, and 1.4% for answer generation. The
624 latency distribution indicates that memory-building operations such as claim processing, trajectory
625 organization, and wiki construction are the main runtime bottlenecks. This suggests that future
626 efficiency improvements should prioritize caching, batching, and replacing some LLM-based memory-
627 construction steps with cheaper deterministic or distilled components.

628 Table 6 summarizes how memory size grows across dialogue samples. On average, each dialogue
629 contains 588.2 raw messages, which TrajWiki organizes into 128.6 trajectories, 294.3 snapshots,

Token Usage by Cost Phase

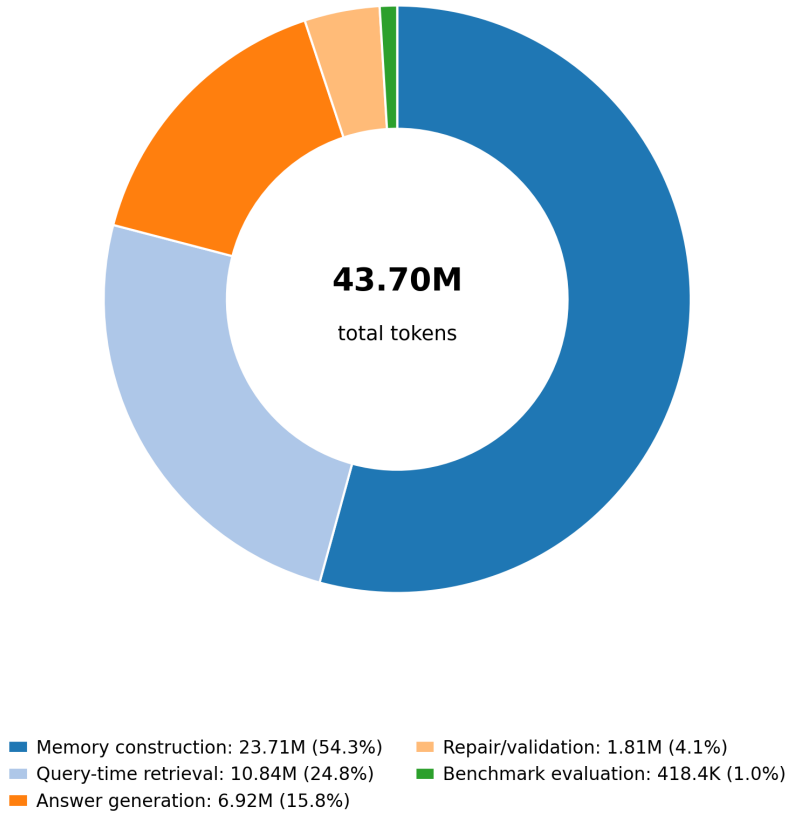


Figure 6: Token usage by cost phase. Token usage is dominated by memory construction and query-time retrieval, while benchmark-only evaluation contributes less than 1% of total measured tokens.

3409.2 claims, and 42.5 wiki pages. Figure 8 provides the same view across individual dialogue samples on a log scale. The figure shows that claims are the largest memory object class, while wiki pages remain much smaller in number than trajectories and snapshots. This supports the role of the Memory Wiki as a compact routing layer over a much larger provenance-rich memory store.

Figure 9 shows the query-time effect of this organization. Direct trajectory retrieval would consider 130.4 trajectories per query on average, whereas wiki routing reduces the candidate universe to 55.6 trajectories, a $2.35\times$ reduction before final top- k selection. This reduction is important because query-time retrieval otherwise scales with the full trajectory store. The wiki layer therefore provides a concrete scalability benefit even though the current implementation remains token-intensive.

Overall, this analysis shows that TrajWiki is currently a computation-heavy, provenance-oriented memory system rather than a token-minimizing retrieval baseline. Its main efficiency benefit in this implementation is search-space reduction through persistent wiki routing. Future cost optimization should focus on reducing memory-construction latency, caching reusable retrieval computations, and replacing some repeated LLM-based organization steps with cheaper deterministic or distilled alternatives.

Provider Latency by Cost Phase

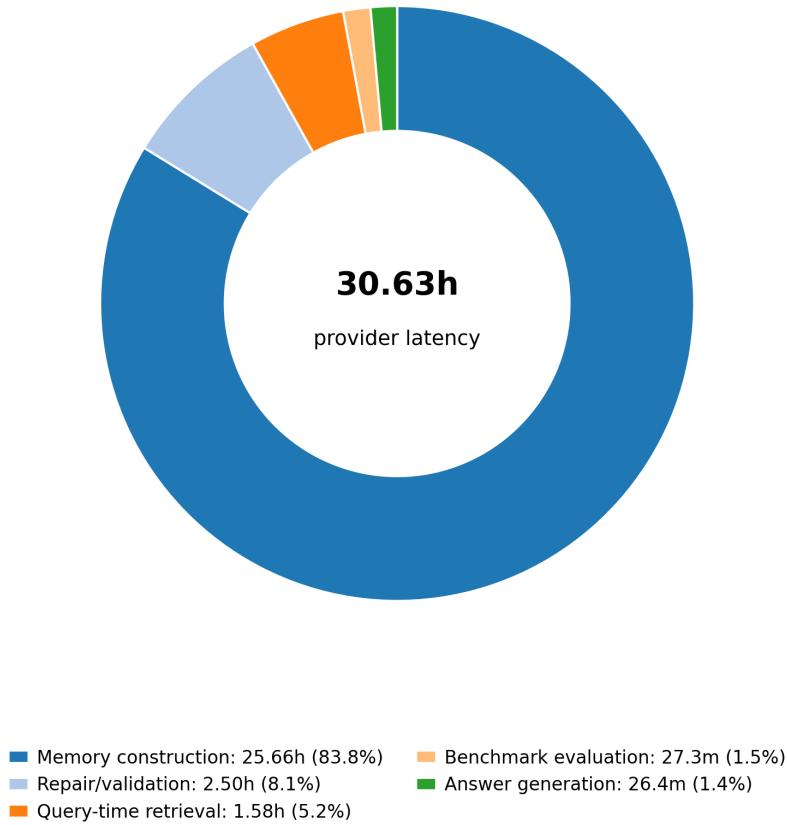


Figure 7: Provider-side latency by cost phase. Accumulated provider latency is primarily spent on memory construction, including extraction, trajectory organization, claim processing, and wiki compilation.

Table 6: Memory size and candidate scaling. Values are averaged across LoCoMo dialogue samples or evaluated queries.

Quantity	Mean	Min	Max
Raw messages per sample	588.2	369	689
Raw memory tokens per sample	19.9K	12.0K	23.9K
Trajectories per sample	128.6	43	188
Snapshots per sample	294.3	185	344
Claims per sample	3409.2	2613	4483
Wiki pages per sample	42.5	30	59
Direct trajectory universe per query	130.4	43	188
Wiki-routed trajectory universe per query	55.6	27	66
Selected trajectories per query	15.0	15	15
Selected snapshots per query	28.3	16	30
Selected source messages per query	74.2	29	150

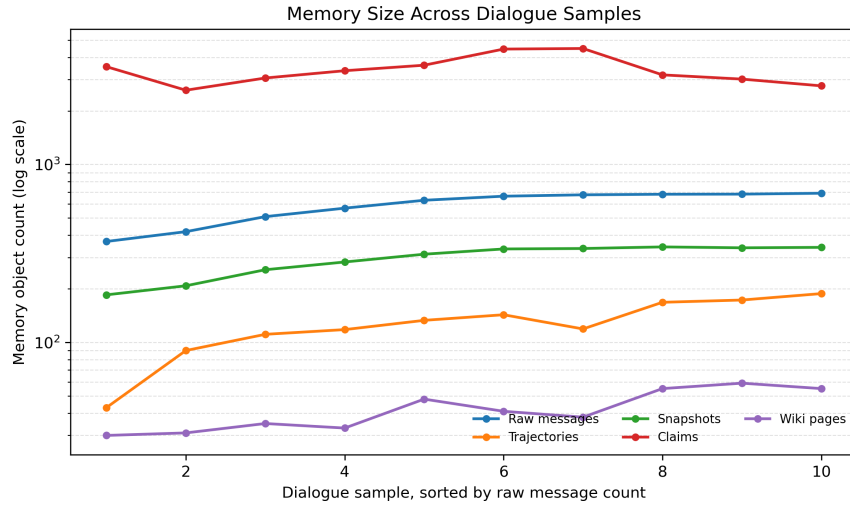


Figure 8: Memory size across dialogue samples. Counts are shown on a log scale. TrajWiki stores raw messages, trajectories, snapshots, claims, and wiki pages as separate memory objects.

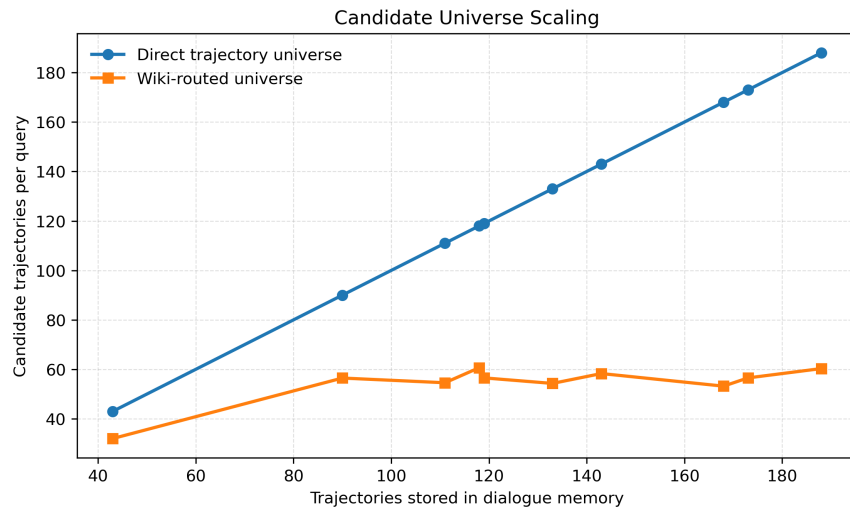


Figure 9: Candidate universe scaling. Wiki routing keeps the trajectory candidate universe substantially smaller than direct retrieval over all stored trajectories.

Table 7: Answer-support diagnostics on LoCoMo. Metrics are offline proxies computed from recorded support references and gold evidence, not human-verified hallucination labels.

Diagnostic	Value
Evaluated queries	282
Source-supported answer proxy	0.390
Unsupported-answer risk proxy	0.610
Mean retrieved gold-reference coverage	0.615
Mean support gold-reference coverage	0.332
Invalid supporting-reference rate	0.216
Unsupported extra-item count	30
Unsupported count-error count	3
Support refs with no gold overlap	78

Table 8: Failure-localization proxy distribution. The table reports automatically assigned diagnostic stages; no human failure-stage labels are used.

Diagnostic Stage	Queries	Rate
Unsupported overgeneration	133	0.472
Answer synthesis error	76	0.270
Trajectory selection miss	28	0.099
Page routing miss	12	0.043
Grounding miss	1	0.004
Correct	32	0.113

645 B.4 Limitations and Diagnostic Visibility

646 We provide additional diagnostics for the limitations discussed in Section 4.8. These diagnostics
647 are computed offline from the LoCoMo run with GPT-4o-mini. They do not introduce extra LLM
648 calls and should be interpreted as observable proxies rather than human-verified audit accuracy. In
649 particular, we do not report true conflict detection accuracy, obsolete-claim handling accuracy, or
650 human auditor time/error rate without external labels.

651 **Source-support and unsupported-answer diagnostics.** Table 7 summarizes answer-support diag-
652 nostics. TrajWiki records supporting references for final answers and compares them with gold source
653 references. The source-supported answer proxy is 0.390, while the unsupported-answer risk proxy is
654 0.610. The mean retrieved gold-reference coverage is 0.615, but the mean support gold-reference
655 coverage is 0.332, indicating that gold evidence is often retrieved but not always preserved as final
656 answer support. This supports the diagnostic value of TrajWiki: it can distinguish retrieval availability
657 from answer-stage grounding failures.

658 **Failure localization.** Table 8 reports the distribution of automatically localized failure stages.
659 These stages are derived from recorded provenance, retrieval metadata, gold evidence coverage, and
660 answer-support diagnostics. They are useful for error analysis, but should not be read as failure-
661 localization accuracy because no human stage labels are provided. The largest category is unsupported
662 overgeneration, followed by answer synthesis error. This indicates that many failures occur after
663 relevant evidence is available, rather than only at the retrieval stage.

664 **Conflict, obsolete-claim, and audit-packet diagnostics.** TrajWiki also records claim lifecycle state
665 and compact audit packets. In this run, conflict-related context was observable for 0.535 of queries,
666 deprecated claims were suppressed in rendered context with rate 1.000, and the deprecated-leakage
667 proxy was 0.128. These numbers show that TrajWiki exposes conflict and obsolete-claim signals, but
668 they are not substitutes for labeled conflict or obsolete-handling accuracy.

669 The audit packet provides a compact unit for human or post-hoc inspection. Compared with a
670 full-context audit proxy, TrajWiki reduces the mean audit source count from 594.7 to 195.1 and the
671 mean audit packet size from 20.33K to 11.34K estimated tokens. This suggests that the provenance
672 structure can make post-hoc inspection more focused, although actual human audit time and error
673 rate remain future work.

Table 9: Auditability diagnostics. Conflict and obsolete-claim metrics are observable proxies. Audit-packet size measures the compactness of evidence made available for inspection.

Diagnostic	Value
Conflict exposed rate	0.535
Deprecated-claim suppression rate	1.000
Deprecated leakage proxy	0.128
Obsolete-answer risk proxy	0.128
TrajWiki mean audit source count	195.1
Full-context proxy source count	594.7
TrajWiki mean audit packet tokens	11.34K
Full-context proxy audit tokens	20.33K
TrajWiki mean audit claim count	315.2

Table 10: Fallback and repair diagnostics on LoCoMo with GPT-4o-mini. Extra tokens and calls quantify the overhead introduced by fallback, repair, and deterministic validation paths.

Stage	Events	Fallbacks	Repairs	Risk Events	Extra Tokens
Answer synthesis / validation	554	0	69	330	25.10M
Repair phase	564	0	1408	158	1.84M
Semantic metrics	848	1	0	1	0.00M
Unknown fallback events	156	7	0	0	0.00M
Total	2122	8	1477	489	26.93M

Fallback and repair overhead. A separate limitation is structured-output dependence. Table 10 summarizes fallback and repair diagnostics. Even with GPT-4o-mini, TrajWiki records substantial repair activity, especially around answer synthesis, validation, and claim preservation. These paths are useful for robustness and diagnosis, but they increase cost and latency. For weaker or non-instruction-tuned models, this dependency is likely to become more severe.

The ledger reports approximately 2,562 extra provider calls, 26.93M extra tokens, and 23.90M ms of accumulated provider-side latency from fallback and repair-related paths. These overheads should not be interpreted as end-to-end failures, but they show that reliable structured generation is an operational requirement for TrajWiki.

Case study: summary-mediated salience drift. A concrete example is query conv-26_qa_24:

Question: What does Melanie do to destress?

Gold answer: Running, pottery

Model answer: Melanie engages in various self-care activities to destress, including running, reading, playing the violin, and prioritizing self-care.

The raw evidence supports both required items: D5 : 4 states that Melanie signed up for a pottery class and described it as “therapy for me,” while D7 : 22 states that she had been running farther to de-stress. This was not a pure retrieval miss: the relevant evidence was present in the memory store. The failure occurred because the pottery fact became less salient through snapshot, trajectory, and wiki-level abstraction. The fact survived in source-linked metadata, but it was not foregrounded strongly enough during final answer synthesis.

This case illustrates the main limitation of provenance-aware summarization. Source links make the failure traceable, but they do not guarantee that every source-specific fact remains salient after repeated abstraction. Future work should combine stricter constrained decoding, periodic source-summary consistency audits, and final-answer verification against raw source evidence rather than generated summaries alone.

C Prompts

This appendix lists the main prompts used by TrajWiki. For tasks using structured output, the prompt is paired with a schema through the provider’s structured-output interface; we show the natural-language instruction portion here.

C.1 Memory Construction Prompts

Episodic Claim Extraction.

```
TASK=EPISODIC_CLAIM_TEXT_EXTRACT_STRUCTURED
Rewrite the current exchange into readable, source-grounded atomic claim sentences.

Return structured data only through the provider's structured output channel.

Rules:
- Use only facts explicitly stated in the raw messages.
- Resolve speaker names from the raw source lines. If the raw line says "(Melanie): my son", write "Melanie's son", not "Caroline's son", "the user", or "the assistant".
- claim text must be a complete readable factual sentence, not a transcript fragment or broad summary.
- For list-like utterances, include item-level claims for every stated item.
- Preserve exact raw surface phrases for names, titles, places, counts, instruments, events, and items.
- Do not collapse books, instruments, cities, places, activities, events, painted objects, or counts into umbrella-only claims.
- Do not replace concrete raw noun phrases with broader categories: if raw says "sunset", keep "sunset", not only "nature-inspired artwork".
- You may include broader context, but the original source phrase must remain visible in at least one claim.
- supporting_quote must be copied from the raw source message and must support the claim.
```

Claim Signal Extraction.

```
TASK=CLAIM_SIGNAL_EXTRACT_STRUCTURED
Extract validated retrieval and display signals from readable claims and source snippets.

Return structured data only through the provider's structured output channel.

Rules:
- Every surface/value/value_span/display value must be copied from a supplied claim or source snippet.
- Do not invent normalized names or semantic replacements.
- Do not output transcript fragments, greetings, acknowledgements, or sentence opener fragments.
- Prefer concise useful retrieval/display signals over exhaustive noisy spans.
```

Trajectory Matching.

```
TASK=TRAJECTORY_MATCH_STRUCTURED
Decide whether the new memory continues one of the shortlisted candidate trajectories.

Return structured data only through the provider's structured output channel.

Rules:
```

```

758 - Choose CONTINUE only if the new memory clearly belongs to the same evolving
759 trajectory as one candidate.
760 - Each candidate includes a trajectory summary for the whole thread and a latest
761 update note for only the newest step.
762 - Use the trajectory summary as the primary signal; use the latest update note
763 only as recent supporting context.
764 - Do not require the new memory to resemble only the latest update if the broader
765 trajectory summary clearly matches.
766 - Same person, broad topic, or generic words such as community/support/art are not
767 enough to CONTINUE.
768 - CONTINUE requires the same evolving event, project, item, place, count, or fact
769 thread.
770 - Choose NEW when the memory is about a different item, place, event, count, or
771 topic cluster under the same person unless it explicitly updates the existing
772 thread.
773 - selected_candidate must be null when decision is NEW.
774 - selected_candidate must be one of the provided candidate labels when decision is
775 CONTINUE.
776 - rationale should briefly explain the shared evolving trajectory or why no
777 candidate fits.
778 - Do not say candidates are merely related or similar.
779

```

781 C.2 Memory Wiki and Retrieval Prompts

782 Trajectory Retrieval Summary.

```

783
784 TASK=TRAJECTORY_RETRIEVAL_SUMMARY
785 Create a markdown retrieval summary for ONE episodic trajectory using only the
786 supplied grounded internal memory state.
787
788 Goal:
789 - produce a retrieval-oriented synopsis for coarse trajectory selection
790 - preserve exact names, items, and specific facts
791 - keep older but query-relevant facts visible even when later snapshots shift
792 topic
793 - surface uncertainty or conflicts explicitly instead of hiding them
794
795 Required headings:
796 ## Profile / Stable Facts
797 ## Item Sets / Named Entities
798 ## Relations / Temporal Updates
799 ## Conflicts / Uncertainty
800
801 Rules:
802 1. Use only the supplied claims, facets, exact terms, and recent snapshot notes.
803 2. Preserve exact titles, books, recipes, instruments, symbols, places, and named
804 items.
805 3. Keep list items explicit; do not collapse them into broad themes.
806 4. Do not let the newest update erase older item sets, places, events, dates, or
807 counts from the trajectory.
808 5. Do not replace concrete source phrases with broad categories: keep "sunset"
809 visible even if you also mention "artwork".
810 6. Mention contradictory or uncertain facts under Conflicts / Uncertainty instead
811 of resolving them yourself.
812 7. Do not add commentary, preamble, or conclusions outside the markdown sections.
813 8. Exclude filler words, acknowledgements, discourse markers, and casual fragments
814 from the useful retrieval content.
815
816

```

818 Wiki Page Planning.


```

820
821 TASK=WIKI_PAGE_PLAN
822 Plan a compact sample-level wiki graph from the provided candidate seed manifest.
823
824 Return markdown only with the exact heading:
825 ## Pages
826
827 Then emit one bullet per page:
828 - page_type=<index/entity/topic/inventory> | title=<title> | slug=<slug> |
829   trajectories=<comma-separated trajectory ids> | entities=<comma-separated
830   entities> | links=<comma-separated page slugs>
831
832 Rules:
833 1. Always include exactly one index page.
834 2. Treat each input block as a candidate seed that already groups related
835   trajectories; refine or select from those seeds instead of inventing broad new
836   groupings.
837 3. Create entity pages when a person/entity appears across multiple trajectories.
838 4. Create inventory pages for list/count/item-heavy facts such as activities,
839   books, recipes, instruments, symbols, places, wins, dogs, or repeated events.
840 5. Create topic pages only when they add navigation value beyond entity or
841   inventory pages.
842 6. Keep pages compact and specific; avoid broad pages that overlap heavily with
843   existing entity or inventory coverage.
844 7. Prefer explicit evidence coverage over narrative grouping.
845 8. Use trajectory historical evidence cards as first-class evidence; do not rely
846   only on a trajectory's latest summary.
847 9. If a card exposes a specific item, place, event, count, or source anchor, cover
848   it in a non-index page whenever possible.
849

```

851 Wiki Page Compilation.

```

852
853 TASK=WIKI_PAGE_COMPILE
854 Compile one wiki page from the supplied page metadata, representative trajectory
855   summaries, and historical trajectory evidence cards.
856
857
858 Return markdown only with these headings:
859 ## Overview
860 ## Key Facts
861 ## Items / Counts
862 ## Linked Trajectories
863 ## Conflicts / Uncertainty
864
865 Rules:
866 1. Use only supplied linked trajectory ids, dominant entities, exact terms, facet
867   values, representative trajectory summaries, and historical evidence cards.
868 2. Keep exact names, items, places, counts, and titles.
869 3. For inventory pages, favor explicit item lists over prose.
870 4. Preserve enumerated items and counts explicitly; do not collapse them into
871   broad narrative summaries.
872 5. Do not replace a concrete source phrase with a broader category: if an evidence
873   card contains "sunset", keep "sunset", not only "nature-inspired artwork".
874 6. Do not add speculative links or inferred facts not stated in the supplied
875   inputs.
876 7. In Items / Counts, use only readable display items, readable counts, and
877   readable key facts.
878 8. If a latest summary and historical card emphasize different facts, present the
879   historical facts in their own concrete bullets instead of dropping them.
880 9. Do not dump Dominant exact terms verbatim; ignore any low-confidence or
881   fragment-like signal.
882 10. Do not write placeholder descriptions such as "Not provided", "Unknown", "N/A",
883     "None provided", "No specific key facts", or "No explicit items".

```

884 11. Describe a linked trajectory only when its representative summary or evidence
 885 card is supplied; otherwise leave the trajectory id without a fabricated
 886 description.
 887 12. The system may rewrite the final Linked Trajectories section deterministically,
 888 so keep that section concise.
 889

891 C.3 Answer Generation and Evaluation Prompts

892 LoCoMo Evidence Synthesis.

893
 894 TASK=LoCoMo_ANSWER_EVIDENCE_SYNTHESIS
 895 Legacy structured LoCoMo evidence synthesis.
 896
 897 Input contains a LoCoMo question followed by retrieved memory/wiki/source evidence.
 898
 899
 900
 901 Rules:
 902 1. Use only retrieved evidence. If it cannot answer the question, set can_answer=
 903 false, leave final_answer empty, and explain the gap in abstain_reason.
 904 2. Preserve exact source terms for names, titles, places, dates, counts,
 905 activities, and items.
 906 3. supporting_source_refs must use only visible refs such as D8:4, and each ref
 907 must support the requested answer family.
 908 4. Count distinct completed events only. Exclude future plans, intentions,
 909 reactions, duplicate mentions, general hobbies, and uncertain candidates.
 910 5. Count final_answer must be natural language, not a bare number; use an exact
 911 count only when complete evidence supports it, otherwise state a retrieved-
 912 evidence lower bound.
 913 6. Date/time answers must use source line date=... fields, snapshot Timestamp
 914 lines, and the ## Temporal Anchors block; resolve terms such as "yesterday" to
 915 the grounded date when available.
 916 7. For lists/events/preferences, include every in-scope source-backed specific
 917 item and avoid scope-mismatched extras.
 918 8. For bridge/alias facts, answer with the concrete source-backed value, e.g. "
 919 West County" rather than "old area".
 920
 921 Return structured data only. If structured output is unavailable, return only a
 922 JSON object matching the same fields.
 923

925 LoCoMo Judge.

926
 927 Evaluate the candidate answer against the question and gold answer.
 928
 929
 930 Equivalence Policy:
 931 - Allow semantic equivalence, near-synonyms, ordering/format variants, harmless
 932 modifiers, and concise or longer phrasing with the same meaning.
 933 - Accept date variants for the same time point, e.g. "May 7th" and "7 May"; an
 934 unanchored relative date such as "last Tuesday" is PARTIAL, and a clearly wrong
 935 date is INCORRECT.
 936 - Treat source-backed aliases as covered: "LGBTQ support group" covers "support
 937 group"; "LGBTQ pride parade" covers "pride parade"; "shared her journey at a
 938 school event and encouraged students" covers "school speech"; "mentorship
 939 program for LGBTQ youth" covers "mentoring program".
 940 - Harmless same-category extras are allowed unless they change the requested scope,
 941 contradict the gold answer, or replace a required item.
 942
 943 Strict Cases:
 944 - Counts are strict: "Twice", "2 times", and "two confirmed rejections" are
 945 equivalent, but an unqualified wrong count is INCORRECT.

946 - A retrieved-evidence lower bound such as "The retrieved evidence confirms one
947 rejection" is PARTIAL when the gold requires more.
948 - Missing a required list item is PARTIAL; for example, omits "school speech" from
949 "pride parade, school speech, support group".
950 - Wrong entity, polarity, count, time, or place is INCORRECT.
951
952 Verdicts:
953 - CORRECT: all required gold facts/items are covered and no contradiction changes
954 the answer.
955 - PARTIAL: at least one required fact is covered, but required content is missing
956 or qualified as a lower bound.
957 - INCORRECT: no required fact matches, or the answer reverses/changes the core
958 meaning.
959

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1088 **6. Experimental setting/details**

1089 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-

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- 1098 material.

1099

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1101 Question: Does the paper report error bars suitably and correctly defined or other appropriate

1102 information about the statistical significance of the experiments?

1103 Answer: [No]

1104 Justification: The paper reports point estimates for the main experimental results, but

1105 does not currently include error bars, confidence intervals, or statistical significance tests.

1106 Repeated full LLM evaluations are costly due to API usage, so we did not run multiple

1107 independent trials or report uncertainty estimates in the current version.

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